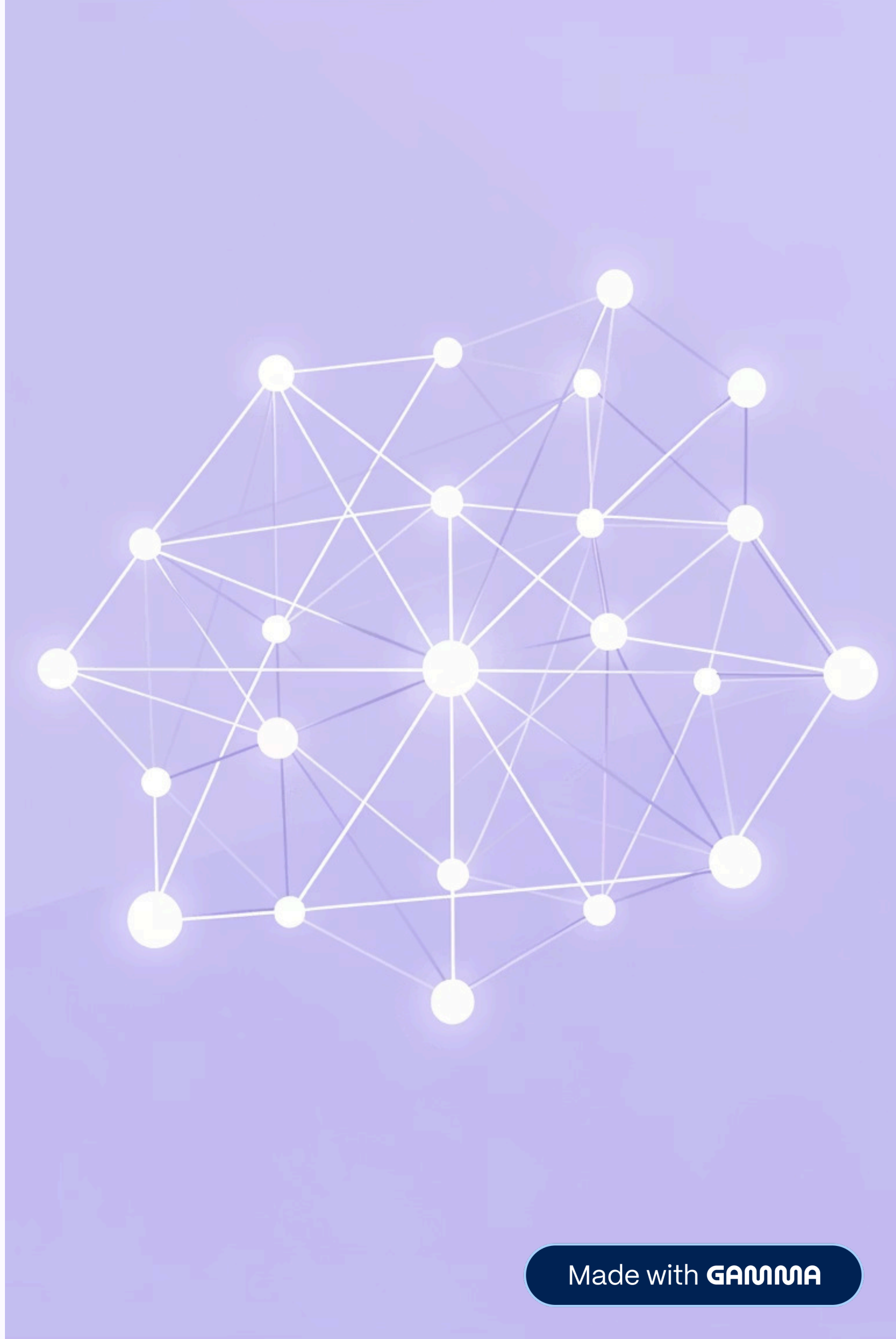
Depth First Search (DFS) & Breadth First Search (BFS)

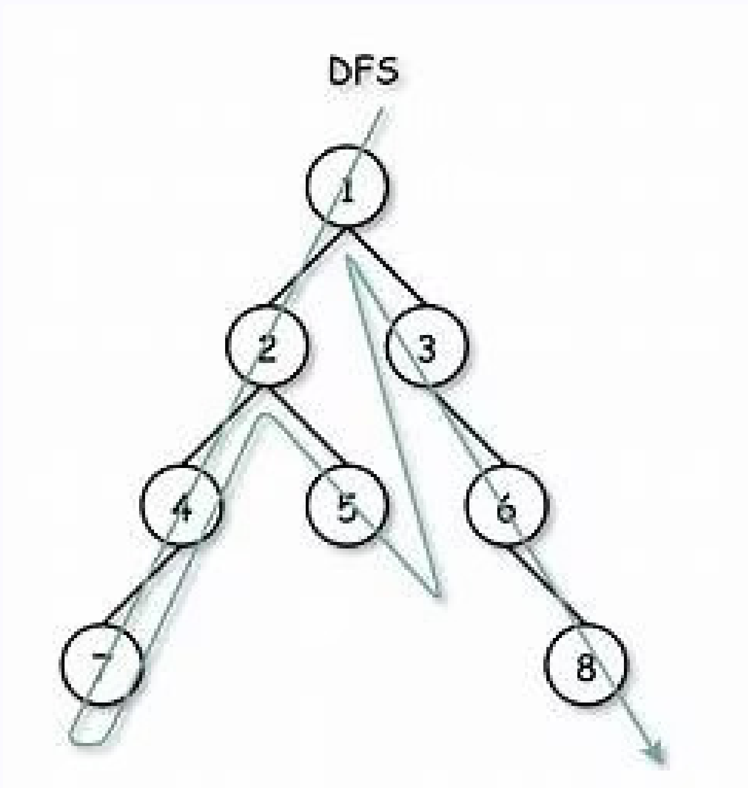


What Are DFS and BFS?

1

Depth First Search (DFS)

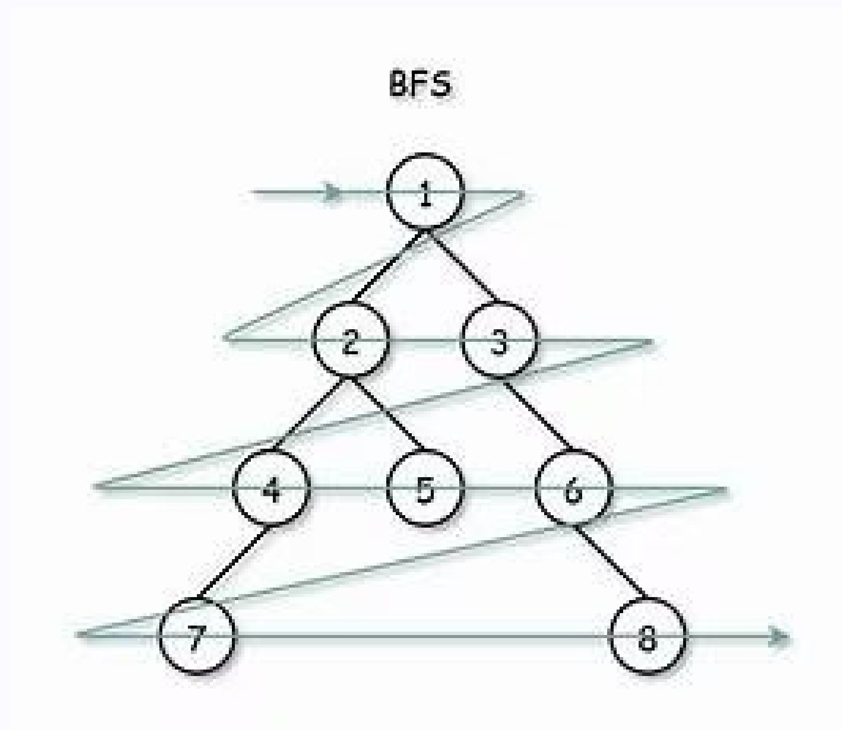
DFS explores as deep as possible along each branch before backtracking. It can be implemented using a stack data structure or implicitly through recursion (call stack).



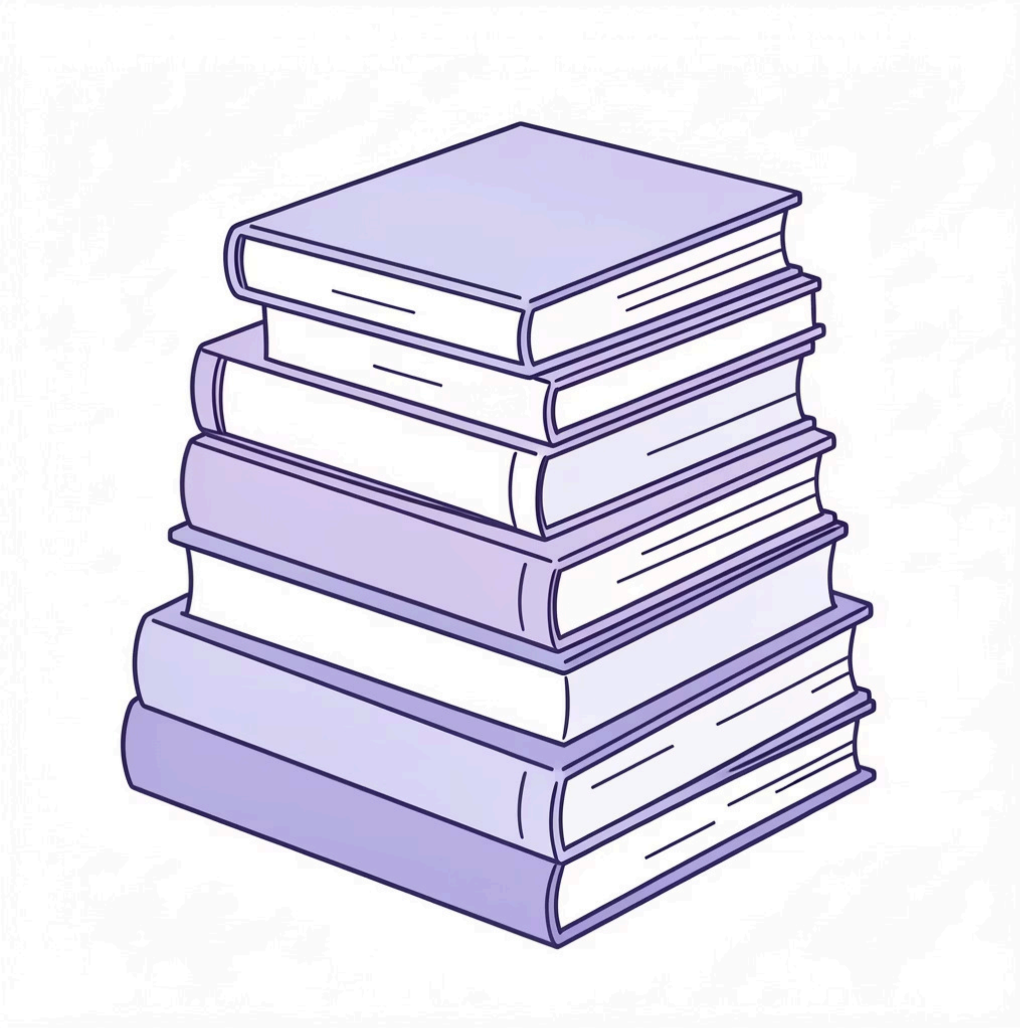
2

Breadth First Search (BFS)

BFS systematically explores all neighbors at the current depth level before moving on to nodes at the next depth level. It utilizes a queue and is ideal for finding the shortest path in unweighted graphs.



Key Differences Between DFS and BFS



DFS: Depth-Oriented

- **Structure:** Uses a stack (LIFO - Last-In, First-Out) for node management.

DFS (Depth First Search)
Uses **stack / recursion**
Traverses **depth-wise**
Goes deep before backtracking
Uses less memory
Does **not** find shortest path
Suitable for backtracking

BFS: Level-Oriented

- **Structure:** Employs a queue (FIFO - First-In, First-Out) to process nodes.

BFS (Breadth First Search)
Uses **queue**
Traverses **level-wise**
Visits all neighbors first
Uses more memory
Finds **shortest path**
Suitable for shortest distance

Applications

- **Applications:** Excellent for exhaustive path exploration, detecting cycles in graphs, and solving problems that involve backtracking, such as puzzles and mazes.

- **Applications:** Best suited for finding the shortest path between two nodes in unweighted graphs, network broadcasting, and exploring nodes layer by layer.

Advantages and Dissadvantages of DFS

Advantages: Low Memory Usage: Uses less memory (stack depth) compared to BFS.

Good for Deep Solutions: Efficient for finding solutions that are far from the source.

Applications: Maze solving, cycle detection, puzzle games.

Disadvantages: Not Shortest Path: Doesn't guarantee the shortest path.

Infinite Loops: Can get stuck in infinite loops in graphs with cycles (needs visited tracking).

Completeness Issues: May not find a solution if it goes down an infinitely deep path first.

Advantages and Dissadvantages of BFS

Advantages:

Shortest Path: Guarantees finding the shortest path in unweighted graphs.

Completeness: Finds a solution if one exists.

Good for Shallow Solutions: Efficient for finding solutions close to the source.

Applications: Social networks, GPS, broadcasting.

Disadvantages:

High Memory Usage: Can use significant memory for wide graphs, storing all nodes at a level.

Slower for Deep Solutions: Can be slower than DFS if the solution is deep in the graph.

Thank You...